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Describe how **one** amino acid is added to a polypeptide that is being formed at a ribosome during translation.

[3 marks]

Question 3 continues on the next page

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Table 2 shows:

- mRNA codons and the amino acid coded for by each codon
- the type of bond formed by the R group of some of the amino acids.

Table 2

First base	Second base				Third base
	U	C	A	G	
U	Phe	Ser	Tyr	Cys	U
	Leu		Stop	Stop Trp	C A G
C	Leu	Pro	His	Arg	U
			Gln		C A G
A	Ile	Thr	Asn	Ser	U
	Met		Lys	Arg	C A G
G	Val	Ala	Asp	Gly	U
			Glu		C A G

Key to the type of bond formed by the R group of each amino acid

☐ Hydrogen bonds
 ☐ Ionic bonds
 ☐ Disulfide bridges

0 3 . 2

Crystallin is a structural protein found in the human eye. An inherited disease that leads to blindness is caused by changes in properties of crystallin. The replacement of the amino acid Arg with the amino acid Gly causes these changes.

Use information in **Table 2** to suggest why this amino acid replacement changes the properties of crystallin.

[2 marks]

0 3 . 3

The amino acid replacement of Arg with Gly is caused by a single base substitution mutation in the DNA. The non-mutant DNA triplet is TCC.

Complete **Table 3**.

Give:

- the mRNA codon complementary to the non-mutant DNA triplet
- the mutated mRNA codon that could cause the change from Arg to Gly in the crystallin protein
- the DNA triplet complementary to this mutated mRNA codon.

[2 marks]

Table 3

mRNA codon for the non-mutant triplet	
Mutated mRNA codon	
Mutated DNA triplet	

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Turn over for the next question

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